

Align-Extract-Score: Graded Relevance of Arabic Essays

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Abstract

Relevance is a central component of Automated Essay Scoring (AES), as it measures how well a student essay addresses a given writing prompt. In this work, we propose AESER, a novel approach to address the problem of graded relevance scoring in Arabic cross-prompt AES. AESER adapts the given prompt more towards the writing space of essays and computes token-level prompt-essay similarity matrices from contextual embeddings. These matrices are processed by a convolutional neural network to predict the relevance score. Experiments on a large Arabic AES dataset show that AESER outperforms strong baselines, establishing a new state-of-the-art performance.

1 Introduction

Automated Essay Scoring (AES) assigns quality scores to essays and offers clear benefits for large-scale assessments where manual grading is costly. A key component of AES is trait-level scoring, which assesses specific aspects of writing, e.g., organization, style, or relevance, to provide more targeted feedback.

The relevance trait, commonly referred to as *prompt adherence* (Persing and Ng, 2014; Mathias and Bhattacharyya, 2018), measures how well an essay addresses the given writing prompt. The prompt defines both the writing task and the topic that the student is expected to address. Relevance is particularly important because it reflects the student’s ability to correctly interpret the topic and maintain focus throughout the essay. It also determines the validity of other traits; an off-topic essay cannot be considered high quality, regardless of its structure or fluency. Considering the importance of this trait, in this work we focus on predicting *graded* (i.e., multi-level) *relevance* of essays given a writing prompt. We address this problem in the *cross-prompt* setting, where models are evaluated

on writing prompts that were *unseen* during training, which reflects realistic educational scenarios where instructors regularly introduce new writing tasks for which no annotated essays are available.

Most of the existing trait-level AES literature focuses on simultaneous scoring across multiple traits, including relevance, but with no special attention given to the particular notion or nature of relevance (Ridley et al., 2021; Do et al., 2023; Chen and Li, 2023; Eltanbouly et al., 2025). Only a few studies directly target the relevance trait (Albatarni et al., 2024). However, the field remains largely English-centric, and *Arabic* relevance scoring is still scarce, typically limited to prompt-specific or binary formulations (Bashendy et al., 2024; Machhout and Zribi, 2024; Qwaider et al., 2025). We address this gap by targeting *graded*, *cross-prompt* relevance scoring for Arabic essays.

To this end, we propose AESER, an Align-Extract-Score approach for Essay Relevance.¹ Our approach *aligns* the given writing prompt to the scope of a written essay by using a large language model (LLM) to “rewrite” it into one or more essays that reflect the prompt’s topic at different relevance levels. An encoder then *extracts* a token-wise similarity matrix between the transformed prompt and the essay, explicitly capturing fine-grained prompt–essay interactions.

Finally, inspired by Pang et al. (2016), we *score* the essay by treating the similarity matrix as a grayscale image passed to a convolutional neural network (CNN), framing similarity patterns in a visual representation for graded relevance scoring. Experiments on LAILA, a recently released large Arabic AES dataset (Bashendy et al., 2026), show that AESER establishes a new state-of-the-art (SOTA) over strong baselines.

Overall, our contributions are three-fold:

1. We propose AESER, a novel interaction-

¹Pronounced as “ACER” with a different spelling.

based relevance scoring framework that uses token-level similarity matrices to capture fine-grained prompt–essay relationships.

2. We show that AESER achieves a new SOTA on LAILA dataset, surpassing all baselines by substantial margins and remaining robust to the choice of LLM.
3. We release our code² for reproducibility and to support future work on AES relevance, and Arabic in particular.

2 Related Work

Our work brings together two lines of research, trait-level cross-prompt AES with the focus on relevance trait modeling, and Arabic AES.

Trait-level Cross-Prompt AES Early cross-prompt work focused on scoring essays *holistically* with shared neural architectures (Jin et al., 2018; Li et al., 2020; Ridley et al., 2020), after which Ridley et al. (2021) introduced *trait-level* cross-prompt scoring, combining shared low-level layers for generalization with prompt-specific high-level layers. Later models kept this hierarchical design and added prompt-aware attention (Do et al., 2023), contrastive prompt-mapping (Chen and Li, 2023) and meta-learning training (Chen and Li, 2024). Eltanbouly et al. (2025) combined hand-crafted features with trait-specific LLM-derived features. Throughout this line of work, relevance is scored jointly with other traits and is not modeled explicitly, despite its importance for essay quality.

Relevance was incorporated in earlier work to support holistic scoring through classic IR signals such as bag-of-words similarity (Rei and Cummins, 2016) and query expansion (Cummins et al., 2016). Albatarni et al. (2024) addressed graded relevance in English essays by embedding essays with an unsupervised dense encoder and scoring each by its nearest relevance centroid (1-NN).

Arabic AES The large majority of the work was English-centric, motivated by the availability of English datasets (Li and Ng, 2024), with little work on other languages including Arabic (Bashendy et al., 2024). Indeed, Arabic AES faces major challenges due to data scarcity. Nevertheless, recent efforts have explored Arabic trait-level scoring. Approaches range from hand-crafted feature-based models (Bashendy et al., 2024; Sayed et al., 2025),

to multitask AraBERT fine-tuning with separate regression heads (Bashendy et al., 2025), few-shot LLM prompting with GPT4o (Almarwani et al., 2025) or 10-shot chain-of-thought prompting with GPT-4.1 (Alnajjar et al., 2025), and more recently, Albatarni et al. (2026) proposed MAPLE for Arabic trait scoring using a meta-learning approach. Though performance on relevance in particular remains limited compared to other traits. Studies specifically targeting *Arabic relevance trait scoring* are scarce: Machhout and Zribi (2024) fine-tuned AraBERT in a prompt-specific setting,³ while Qwaider et al. (2025) compared multiple methods on real and synthetic data, finding 2-shot prompting with Falcon and Qwen most effective for *binary* relevance classification rather than graded scoring.

Despite the progress in Arabic AES, relevance scoring has received limited attention and has been studied primarily in prompt-specific or binary settings. We address these gaps by focusing on graded relevance scoring in the cross-prompt setting.

3 Methodology

Relevance scoring task in the context of automated essay scoring aims to assess the topical alignment of a student-written essay with a given writing prompt. Formally, for a prompt p and an essay e , a relevance scoring model maps (p, e) to a score $s \in \{0, 1, \dots, K\}$.

In this section, we present our proposed model, AESER, illustrated in Figure 1. Existing AES relevance models typically use the prompt in its original form alongside the essay, without reformulating it to better align with student responses or modeling fine-grained token-level interactions. In contrast, our method adapts query rewriting from information retrieval to AES and leverages explicit interaction modeling for graded relevance scoring.

AESER consists of three main phases: *Prompt Representation* (subsection 3.1), where the writing prompt is transformed to better align with student essays; *Feature Extraction* (subsection 3.2), where a token-level similarity matrix captures prompt–essay interactions; and *Graded Relevance Scoring* (subsection 3.3), where these signals are aggregated to capture hierarchical patterns from token-level alignments to higher-level structures, producing a normalized relevance score.

²<https://github.com/Maroibo/AESER>

³Where models are trained and evaluated on essays written for the same prompt.

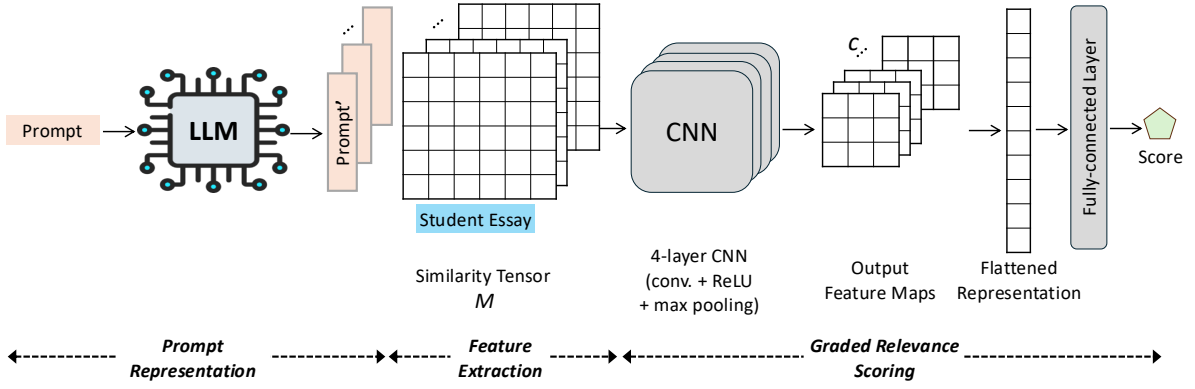


Figure 1: Overview of AESER including prompt representation, feature extraction, and graded relevance scoring.

3.1 Prompt Representation

The writing prompt typically includes *both* the *topic*, which defines the subject the student must address, and the *instructions*, which specify the essay type, word limit, and general structural requirements.⁴

While the topic part defines the scope and the key aspects that the essay should address, the instructions part can *distract* the model’s ability to judge relevance by causing topical drift. Therefore, the topic part of the prompt potentially provides a more reliable basis for relevance assessment, motivating the use of an *instruction-free* prompt that preserves only the core subject.

Nevertheless, an instruction-free prompt remains structurally and length-wise different from the student essays being evaluated, as it represents a concise topic description rather than a complete essay. To address that, we consider prompt representations that resemble student writing. Rewriting the prompt as a *pseudo essay* reflects that relevance manifests in essay-level structure and content, enabling the model to compare texts of similar form. Extending this idea, *rubric guided pseudo essays* incorporate the grading rubric by generating one essay per relevance level, each capturing its expected characteristics and enabling comparison with graded reference representations.

To that end, we examine four different prompt representations:

1. **Original Prompt (p_{orig}):** The prompt is used without any modification, as a baseline representation.
2. **Instruction-free Prompt (p_{topic}):** The prompt

is transformed to retain the topic component, excluding instructional content.

3. **Pseudo Essay (p_{essay}):** The original prompt is transformed into an essay that addresses the given topic.
4. **Rubric-based Pseudo Essays (p_{rubric}):** The prompt is represented as multiple essays, each reflecting a distinct relevance level defined by the rubric.

Each representation strategy produces T text sequences per prompt. For p_{orig} , p_{topic} , and p_{essay} , $T = 1$. For p_{rubric} , one sequence is generated per relevance level, resulting in $T = K$.

We note that in this work, we leveraged the powerful capabilities of LLMs to generate the transformed prompts. The exact instructions prompted to the LLM to generate p_{topic} , p_{essay} , and p_{rubric} prompt representations are depicted in Figure 2.

3.2 Feature Extraction

As relevance depends on how well essay content adheres to the prompt, the feature extraction stage computes a *token-level* similarity tensor to explicitly model prompt–essay alignment. An off-the-shelf encoder E generates contextualized token embeddings for each generated text sequence $p^{(t)}$ and the essay e :

$$\vec{u}_i^{(x)} = E\left(p^{(t)}\right)_i, \quad \vec{v}_j = E(e)_j, \quad (1)$$

where $\vec{u}_i^{(x)}$ is the embedding of the i -th token of the t -th generated text sequence, and $\vec{v}_j$ is the embedding of the j -th essay token.

As illustrated in Figure 1, a similarity tensor $M \in \mathbb{R}^{T \times |p'| \times |e|}$ is then constructed as:

⁴An example of this distinction is shown in Figure 5, where the topic is marked in bold.

English translation. You are an expert in analyzing Arabic texts. Your task is to extract only the main writing topic from the following text, removing all writing instructions and directives. Rephrase the topic clearly and directly. Very important: do not add any comments, questions, or extra explanations. Only the topic.

Text:

(the writing prompt)

The topic only (without any extra comments):

(a) Prompt for the topic representation $\mathbf{p}_{\text{TOPIC}}$. The model is asked to drop the writing instructions and return only the essay topic.

English translation. You are an expert essay writer. Write a coherent, well-organized essay in Arabic that addresses the following writing topic:

(the writing prompt)

The essay:

(b) Prompt for the essay representation $\mathbf{p}_{\text{ESSAY}}$. The model writes a full essay on the topic, which we use as a hypothetical-document expansion of the prompt.

English translation. You are an expert essay writer. Write a coherent, well-organized essay in Arabic that addresses the following writing topic:

(the writing prompt)

Relevance evaluation criteria:

(the rubric describing relevance level ℓ)

The essay:

(c) Prompt for the rubric representation $\mathbf{p}_{\text{RUBRIC}}$. We run this prompt once per relevance level ℓ : the rubric placeholder is filled with the descriptor of level ℓ so the generated essay is written to match that exact level. With K levels this produces K essays per writing prompt, one targeted at each level.

Figure 2: English translated versions of the LLM prompts we passed to generate $\mathbf{p}_{\text{TOPIC}}$ (Figure 2a), $\mathbf{p}_{\text{ESSAY}}$ (Figure 2b), and $\mathbf{p}_{\text{RUBRIC}}$ (Figure 2c). The original prompt is depicted in Figure 3

$$M_{tij} = \phi(\vec{u}_i^{(t)}, \vec{v}_j), \quad (2)$$

where ϕ denotes a vector similarity function (e.g., cosine similarity or dot product).

3.3 Graded Relevance Scoring

The similarity tensor M can be viewed as a stack of 2D interaction signals capturing token-level correspondences between the prompt representation and the essay. Interpreting this tensor as structured spatial input enables a CNN to model local alignment patterns and progressively aggregate them into higher-level interaction features.

Motivated by this intuition, we design a 4-layer convolutional architecture in which kernel sizes increase across layers, and the number of feature maps expands progressively.

Let $L(= 4)$ denote the total number of convolutional layers. We denote by $Z^{(\ell)}$ the output of convolutional layer ℓ , with $Z^{(0)}$ as the input to the first layer. Since $|p'|$ and $|e|$ vary across instances, adaptive max pooling $\mathcal{P}$ is applied to M to obtain a fixed spatial resolution of $S \times S$, resulting in $Z^{(0)} = \mathcal{P}(M) \in \mathbb{R}^{T \times S \times S}$.

The output of each layer, $Z^{(\ell)}$, is computed as:

$$Z^{(\ell)} = \mathcal{P}\left(\sigma\left(W^{(\ell)} * Z^{(\ell-1)} + b^{(\ell)}\right)\right) \quad (3)$$

where $*$ denotes the 2D convolution operator, $W^{(\ell)}$ and $b^{(\ell)}$ are the learnable convolution kernels and biases of layer ℓ , and $\sigma(\cdot)$ is the ReLU activation function. A progressive reduction scheme is employed so that earlier layers retain fine-grained local alignment patterns, whereas deeper layers operate over coarser representations, thereby capturing broader and more global matching structures.

After the last convolutional and pooling layers, the tensor $Z^{(L)}$ is flattened into a vector $\vec{z}$, as shown in the final stage of Figure 1, representing dense relevance features learned by the CNN. At this stage, hand-crafted features can be incorporated by concatenating them with $\vec{z}$. The resulting vector is then passed through a fully connected layer followed by a regression head to predict a continuous relevance score. The score is clipped to the interval $[0, 1]$. During training, the original discrete relevance labels are linearly rescaled to $[0, 1]$, and the model is optimized using mean squared error (MSE) between the predicted score and the scaled target. At inference, the predicted score is mapped back to the original discrete label range to obtain the final integer relevance score.

4 Experimental Setup

This section describes our experimental setup, covering the implementation and hyperparameter tuning of AESER, the dataset, the LLMs used to generate the prompt representations, the evaluation measure, and the baselines.

Prompt Representation Setup We consider three LLMs for generating prompt representations:

- Fanar-27B (hereafter Fanar) is a medium-size Arabic-centric LLM accessed via API.⁵
- Gemma 4 31B (hereafter Gemma) (Google DeepMind, 2026) is a medium-size multilin-

⁵<https://api.fanar.qa/docs>

أنت خبير في تحليل النصوص العربية. مهمتك هي استخراج موضوع الكتابة الأساسي فقط من النص التالي مع إزالة جميع التعليمات الكلاية والتوجيهات. أعد صياغة الموضوع بشكل واضح ومباشر. مهم جداً: لا تضع أي تعليقات أو أسئلة أو شروحات إضافية. فقط الموضوع.
النص:
(نص الموضوع)
الموضوع فقط (بدون أي تعليقات إضافية):

(a) LLM prompt for the topic representation p_{TOPIC} .

أنت كاتب مقالات خبير. اكتب مقالاً متماسكاً ومنظماً باللغة العربية يتناول موضوع الكتابة التالي:
(نص الموضوع)
المقال:

(b) LLM prompt for the essay representation p_{ESSAY} .

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اكتب مقالاً متماسكاً ومنظماً باللغة العربية يتناول موضوع الكتابة التالي:
(نص الموضوع)
معايير التقييم للصلة بالموضوع:
(معايير التصحيح للمستوى المستهدف)
المقال:

(c) LLM prompt for the rubric representation p_{RUBRIC} .

Figure 3: LLM prompts for generating different representation for p_{ORIG} . English translation for each prompt is depicted in Figure 2.

gual open-weights LLM, served with vLLM⁶ for faster inference.

- The closed-source GPT-4o (OpenAI, 2024) is the highest-ranked OpenAI model on the SILMA Arabic LLM Leaderboard at the time of writing.⁷

For all LLMs, the temperature was set to 0.7 and the maximum number of generated tokens (max_tokens) to 1000, with all other hyperparameters left at their default values. The LLM prompts used to generate p_{TOPIC} , p_{ESSAY} , and p_{RUBRIC} are shown in Figure 3a, Figure 3b, and Figure 3c, respectively. For LLM-based representations, we further generate N instances for each text sequence, so that for a prompt representation producing T text sequences, a total of $N \times T$ text sequences are generated. These generated sequences are then added as separate input channels to the CNN.

Implementation, Training, and Hyperparameter Tuning Experiments were conducted on an NVIDIA RTX A6000 GPU. Models were trained

for up to 150 epochs using AdamW with a batch size of 32 and a learning rate of $\eta = 5 \times 10^{-4}$. Early stopping was applied using a patience of 15 epochs based on the development set performance. Token embeddings were obtained using Qwen3-Embedding-0.6B.⁸ We selected this model because it ranks first among models under 1B parameters on the MTEB leaderboard.⁹ For the hand-crafted features, we used the 816 features introduced by Sayed et al. (2025) for Arabic AES, covering surface, readability, lexical, semantic, and syntactic aspects.

The input similarity matrices were pooled to a fixed spatial resolution of $S = 256$. The CNN backbone consisted of four convolutional layers, with $c_{l_1} = 32$ denoting the number of output maps in the first layer. Subsequent layers followed a progressive doubling scheme: $2c_{l_1}$, $4c_{l_1}$, and $8c_{l_1}$. Let $r_{l_1} = 5$ denote the kernel size of the first layer. The kernel sizes of subsequent layers followed an incremental scheme: $r_{l_1} + 2$, $r_{l_1} + 4$, and $r_{l_1} + 6$. Adaptive max pooling was applied after each convolutional layer, with output sizes following a progressive reduction scheme: $o_{l_1} = 256$, $o_{l_2} = o_{l_1}/4$, and $o_{l_{3-4}} = o_{l_1}/16$. The hidden dimension of the fully connected layer denoted by h was initially fixed to 128.

AESER consists of several components that can be tuned, including the prompt representation, the similarity function, the number of retrieved references (N), the inclusion of hand-crafted features, the loss function, and the choice of the generation LLM. To study these components in isolation, we first fix the generation LLM to Fanar and search for the best configuration based on the development set performance.

Specifically, we perform sequential tuning over the components in a fixed order: (1) prompt representation, (2) similarity function, (3) number of generated references (N), (4) inclusion of hand-crafted features, and (5) loss function. For each component, we evaluate all candidate choices while keeping previously selected components fixed, and retain the best-performing option before moving to the next stage. This procedure yields a single optimized configuration, which is then used across all generation LLMs in the subsequent experiments.

⁶<https://github.com/vllm-project/vllm>

⁷<https://huggingface.co/spaces/silma-ai/Arabic-LLM-Leaderboard>

⁸<https://huggingface.co/Qwen/Qwen3-Embedding-0.6B>

⁹<https://huggingface.co/spaces/mteb/leaderboard>

We then tune the hyperparameters $S, r_{\ell_1}, c_{l_1}, o_{l_1}, h$, and η of this configuration separately for each generation LLM on the development set. Specifically, $S \in \{128, 256, 512\}$, $r_{\ell_1} \in \{3, 5, 7, 9\}$, $c_{l_1} \in \{8, 16, 32, 64, 128\}$, $o_{l_1} \in \{128, 256, 512\}$, $h \in \{64, 128, 256\}$, and $\eta \in \{1e^{-5}, 1e^{-4}, 5e^{-4}\}$. Hyperparameters were optimized via Bayesian optimization using the Tree-structured Parzen Estimator (TPE) algorithm (Bergstra et al., 2011), implemented with Optuna’s TPESampler.¹⁰ We conducted 100 trials with 5 startup trials and a fixed random seed. We then evaluate each LLM’s best-tuned model on the test set.

Because most essays in LAILA are fully relevant, we compare class-weighted MSE (WMSE) and focal MSE (FMSE) with standard MSE to test whether reweighting training errors improves relevance scoring. For WMSE, weights were computed using the training split by dividing the number of essays in the most frequent relevance class by the number of essays in each class. Each essay’s squared error was multiplied by its class weight before averaging. FMSE places greater relative emphasis on larger prediction errors by averaging absolute prediction errors raised to $2 + \gamma$, with focusing parameter $\gamma \in \{0.5, 1, 2\}$.

Dataset LAILA dataset is the first large-scale Arabic AES dataset with annotations for seven writing traits including relevance, which is scored on a 0–2 scale (Bashendy et al., 2026). It contains 7,859 essays in response to eight different prompts spanning two types: persuasive and explanatory. The relevance labels are heavily imbalanced toward fully relevant essays: 86% are scored 2 (fully relevant), 6% are scored 1 (partially relevant), and 8% are scored 0 (not relevant). We use the official cross-prompt data split shared by LAILA authors.¹¹

Evaluation Measure We evaluate our approach using Quadratic Weighted Kappa (QWK) (Cohen, 1968), a standard metric in AES. QWK measures agreement between human and system scores while accounting for the severity of disagreements.

Baselines We adopt SOTA models for relevance cross-prompt evaluation: (1) **AraBERT4GR**: Originally proposed for English essay relevance

grading (Albatarni et al., 2024). We adapted it to Arabic by replacing Contriever with AraBERT¹² and tuned all hyperparameters on our training data. (2) **MOOSE**: An adaptation of an English cross-prompt SOTA model based on mixture-of-experts architecture (Chen et al., 2025; Bashendy et al., 2026). (3) **Fanar**^{fs} (5-shots): Second-best model for relevance on LAILA (Bashendy et al., 2026). (4) **Qwen**^{fs} and (5) **Falcon**^{fs} (3-shots): Top-performing models on QAES and ZAEBUC by Qwaider et al. (2025). We reformulated their binary setup into a three-class task aligned with LAILA relevance levels.¹³

5 Experimental Evaluation

This section presents and discusses our results in two parts. We first report the main results that answer our research questions in subsection 5.1. Then, we analyze the model’s performance more closely through an error analysis and a case study in subsection 5.2.

5.1 Main Results

We address three research questions: (**RQ1**) What is the optimal configuration for AESER? We examine different prompt representations, similarity functions, the number of generated instances (N), the inclusion of additional features, and the loss function, (**RQ2**) Which generation LLM performs best for AESER?, and (**RQ3**) How does AESER perform compared to SOTA baselines?

RQ1: AESER Configuration We first examine the different *prompt representations*, depicted in the first section of Table 1. Among them, p_{TOPIC} performed the worst, scoring nearly 7 points below p_{ORIG} , indicating that the discarded information is essential for scoring. p_{ESSAY} yielded only marginal gains over p_{ORIG} despite producing longer prompts; suggesting that length alone is insufficient for improvement. In contrast, p_{RUBRIC} achieved the best performance, with 6 points gain, as it extends p_{ESSAY} with explicit score-level distinctions, offering more informative guidance. Accordingly, p_{RUBRIC} is selected for all subsequent experiments.

We then evaluate the effect of different *feature extraction* setups, including the similarity function

¹⁰<https://optuna.readthedocs.io/en/stable/reference/samplers/generated/optuna.samplers.TPESampler.html>

¹¹<https://gitlab.com/bigirqu/laila>

¹²<https://huggingface.co/aubmindlab/bert-base-arabert>

¹³We keep their prompts and hyperparameters. Few shots sampled randomly; results averaged over 3 runs with different seeds.

Experiment	p	ϕ	N	Features	Loss	QWK _{avg}
Prompt Representation	p_{ORIG}	Cos	1	×	MSE	0.511
	p_{TOPIC}	Cos	1	×	MSE	0.444
	p_{ESSAY}	Cos	1	×	MSE	0.516
	p_{RUBRIC}	Cos	1	×	MSE	0.573
Feature Extraction Setup	p_{RUBRIC}	Dot	1	×	MSE	0.423
	p_{RUBRIC}	Cos	2	×	MSE	0.565
	p_{RUBRIC}	Cos	3	×	MSE	0.550
	p_{RUBRIC}	Cos	4	×	MSE	0.520
	p_{RUBRIC}	Cos	5	×	MSE	0.543
	p_{RUBRIC}	Cos	1	✓	MSE	0.542
	p_{RUBRIC}	Cos	1	×	WMSE	0.438
Loss Function	p_{RUBRIC}	Cos	1	×	FMSE _{0.5}	0.549
	p_{RUBRIC}	Cos	1	×	FMSE ₁	0.507
	p_{RUBRIC}	Cos	1	×	FMSE ₂	0.502

Table 1: AESER performance on LAILA dev set. ϕ is the similarity function, N is the number of generated instances, ‘Features’ indicates hand-crafted features inclusion, and FMSE _{γ} denotes focal MSE with γ .

ϕ , the number of generated instances (N), and the inclusion of hand-crafted features. Results in the second section of Table 1 show that cosine similarity consistently outperformed dot product. Increasing the number of generated instances beyond $N = 1$ did not improve performance; instead, performance degraded as N increased. Incorporating hand-crafted features also did not yield improvements, consistent with prior findings (Bashendy et al., 2026; Albatarni et al., 2026).

We finally examine the loss function in the last section of Table 1. Neither FMSE nor WMSE improved performance over standard MSE. FMSE performance decreased as γ increased, while WMSE resulted in an even larger QWK drop. These results motivate the need for more effective approaches to address class imbalance, which we leave to future work. Overall, the best configuration uses cosine similarity, $N = 1$, and no hand-crafted features, with MSE as the loss function.

RQ2: Best Generation LLM The best configuration identified in RQ1 is applied to different generation LLMs, tuning the hyperparameters separately for each LLM. Table 2 reports AESER on the test set across different generation LLMs. Overall, no single generator consistently dominates across prompts: Fanar leads on P1, P4, P5, and P8, Gemma on P3 and P7, and GPT-4o on P2 and P6. Gemma achieves the best average performance, with Fanar close behind, while GPT-4o trails both by approximately 5 points. These results indicate that Arabic-centric and open multilingual LLMs can match and even outperform the proprietary GPT-4o, suggesting that a strong closed-source

Actual	0	143	228	265
	1	14	133	314
	2	12	396	6354
		0	1	2
		Predicted		

Figure 4: Confusion matrix of AESER_{Gemma}, our best configuration, on the full LAILA test set.

model is not necessary for generating prompt representations in AESER.

RQ3: AESER vs. SOTA Table 2 compares AESER with the baselines. The best, AESER_{Gemma}, reaches an average QWK of 0.517, improving over the strongest baseline, MOOSE, by 11 points (26% relative). AESER also obtains the best score on seven of the eight prompts, with P1 being the only exception. Notably, both AESER_{Fanar} and AESER_{Gemma} show substantial gains on P5 compared to other prompts, improving by 34 and 29 points, respectively, over MOOSE. This improvement can be attributed to the highly specific nature of the prompt topic, “*Qatar’s development and culture*.” Unlike more open-ended prompts such as “*staying up late*,” P5 is grounded in a narrow and concrete topical space, which reduces semantic ambiguity.

Moreover, AESER_{Fanar} performance tends to be stronger on persuasive prompts (P4, P6, and P8), with an average improvement of 14 points over MOOSE compared to explanatory prompts (P1 and P7). This likely reflects the more structured nature of persuasive writing (clear stance, arguments, and conclusion), which promotes greater lexical and discourse-level alignment between generated and student essays. In contrast, as noted in LAILA dataset (Bashendy et al., 2026), explanatory essays have no particular structure, allowing for wider variation in content and organization and making relevance distinctions more challenging.

5.2 Performance Analysis

To understand where AESER succeeds and fails, we examine its predictions in more detail. We first inspect its errors across the three relevance levels through a confusion matrix, and then study two contrasting essays to see how the model judges rel-

Model	P1	P2	P3	P4	P5	P6	P7	P8	Ave.
AraBERT4GR (Albatarni et al., 2024)	0.173	0.396	0.304	0.362	0.186	0.361	0.087	0.307	0.272
Qwen ^{fs(3)} (Qwaider et al., 2025)	0.510	0.133	0.281	0.229	0.446	0.285	0.163	0.255	0.288
Falcon ^{fs(3)} (Qwaider et al., 2025)	0.312	0.110	0.355	0.260	0.335	0.129	0.172	0.101	0.222
Fanar ^{fs(5)} (Bashendy et al., 2026)	0.291	0.476	0.497	0.361	0.478	0.378	0.363	0.424	0.409
MOOSE (Bashendy et al., 2026)	0.119	0.471	0.406	0.506	0.325	0.415	0.593	0.450	0.411
AESER _{Fanar}	0.298	0.472	0.468	0.628	0.669	0.555	0.429	0.598	0.514
AESER _{Gemma}	0.217	0.528	0.533	0.523	0.610	0.553	0.663	0.505	0.517
AESER _{GPT-4o}	0.199	0.531	0.414	0.595	0.393	0.584	0.658	0.362	0.467

Table 2: AESER results on the LAILA test set. AESER_{FANAR}, AESER_{Gemma}, and AESER_{GPT-4o} use FANAR-27B, Gemma 4 31B, and GPT-4o, respectively, to generate the prompt representations. Best score per prompt in **bold**.

Writing prompt (P5, Explanatory).	
تعمل دولة قطر على تحقيق توازن بين تراثها الثقافي والحداثة. اكتب مقالاً مكوناً من ثلاثمائة (300) كلمة توضح فيه السبل والوسائل التي تستخدمها الدولة في تحقيق هذا التوازن، مُراعياً سمات المقال التفسيري، وسلامة اللغة، وموظفاً علامات الترقيم وأدوات الربط بشكلٍ صحيح.	
English translation: The State of Qatar is working to achieve a balance between its cultural heritage and modernity. Write a 300-word essay explaining the methods and means the country uses to achieve this balance, taking into account the features of an expository essay, ensuring linguistic correctness, and using punctuation marks and linking devices properly.	
Essay 50310 (Relevance score: 1, AESER_{Gemma}: 1)	Essay 50124 (Relevance score: 1, AESER_{Gemma}: 2)
السبل و الوسائل التي تستخدمها الدولة في تحقيق هذا التوازن . اولا تحفيز المواطنين و المقيمين في الدولة . ثانيا الحفاظ على الاهمام بالبيئة و نضافتها و نشر اهميتها في المجتمع و الانتباه عليها . ثالثا التعاون فهو مهم في توحيد الدولة و هذا يزيد المحبة في البلد و نشر التعاون و اهمية في الدولة للحفاظ على الاستقرار و تحقيق التوازن في الدولة . رابعا الاهتمام في الرياضة تطور الرياضة يزيد العديد من السكان بسبب حبها للرياضة وهناك الكثير من الرياضات مثل : كرة القدم , كرة السلة , كرة الطائرة ,كرة الامريكية و المزيد من الرياضات . خامسا الامور الدينية يجب علينا نشر قيمنا و ديننا للعالم و ثقافتنا للكسب في الدنيا و الاخرة .	التراث الثقافي والحداثة هي ظواهر طبيعية تحدث مع تغير الاجيال والقرون بين الناس ،وكما انه في الزمن الحالي نرى انه تنتشر الحداثة والتطور في دولتنا قطر، كما ان دولة قطر تفتخر وتحافظ على التراث الثقافي والقطري ،وسأحدثكم في الاسطر التالية عن الوسائل التي تستخدمها دولة قطر في تحقيق هذا التوازن . ومن وسائل تحقيق التوازن ان دولة قطر تحافظ على المباني التراثية و الاحتفاظ ب الادوات التراثية القطرية في المتاحف . وكما ان الشعب القطري يحافظ على العادات والتقاليد مثل : لبس الثوب القطري و رقص العرضة في المناسبات ،وكما انه اعتاد على كرم الضيافة على سبيل المثال : كرم اهل قطر في كأس العالم 2022 ،وكما انهم نشروا الثقافة القطرية في ذلك الزمان . في الختام، يجب علينا الافتخار ونشر ثقافتنا القطرية بين الزوار و الاحتفاظ بكرم دولة قطر.

Figure 5: Two essays from Prompt 5, both labeled as partially relevant (score 1) by the human annotator. The prompt uses bold to mark the topic. AESER_{Gemma} correctly scores the **left** essay, but over-scores the **right** essay.

evance in practice.

Error Analysis Figure 4 shows the confusion matrix of AESER_{Gemma}, our best-performing configuration, over LAILA test set across all prompts. The model is highly accurate on fully relevant essays (score 2), correctly classifying 6,354 of 6,762. It is much weaker on the two minority classes, recovering only 143 of 636 not-relevant essays (score 0) and 133 of 461 partially relevant essays (score 1). These minority essays are mostly over-predicted toward the majority class, with 265 not-relevant and 314 partially relevant essays assigned score 2. This behavior is largely due to the severe class imbalance of the relevance labels (section 4), where the vast majority of essays are fully relevant. AESER’s bias toward the majority class fur-

ther motivates exploring class-balanced training or data augmentation to improve minority-class performance.

Case Study To complement the aggregate analysis, Figure 5 shows two essays written for the same explanatory prompt on Qatar’s efforts to balance cultural heritage and modernity. Both are labeled as partially relevant (1), but they reflect different patterns in how relevance is judged. The first essay (left) is clearly weak in task fulfillment, relying on a fragmented list structure with minimal sentence development and little coherent explanation of how balance is achieved, making its low engagement with the prompt easy to identify. The second essay is more fluent and includes explicit references to culturally salient entities such as “traditional

dance” and “Qatari dress,” which increases its apparent relevance; however, it remains largely descriptive, focusing on heritage display rather than explaining mechanisms of balance, and it omits the modernity aspect. These patterns are expected given the information available to AESER, namely token-level prompt–essay similarity matrices (subsection 3.2). As a result, AESER responds to surface topical cues and shared entities but does not model task completion by design. Consequently, strong topical overlap can raise the predicted relevance even when the essay does not fully address the prompt.

6 Conclusion and Future Work

In this paper, we presented AESER, a framework that formulates relevance assessment of essays as an interaction-based matching task. Combining LLM-based prompt reformulation with token-level similarity matrices and convolutional matching, AESER enables fine-grained modeling of prompt–essay alignment. Our experiments on a large Arabic dataset show that aligning prompt representations with the relevance rubric provides an effective basis for relevance scoring, with AESER outperforming strong baselines by 10 points. Future work will explore more effective approaches to address class imbalance and apply AESER to essays in other languages to evaluate its generalizability.

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7 Limitations

AESER relies on an LLM to generate prompt representations, which introduces an additional requirement compared to feature-based or fine-tuned approaches, and its computational cost increases as the rubric introduces more relevance levels. Despite this, our results indicate that relatively small Arabic-centric or multilingual LLMs are generally sufficient (section 5), which substantially reduces practical cost compared with larger proprietary LLMs. The evaluation is also relatively lim-

ited in scope, as it is conducted on a single Arabic dataset with three relevance levels, leaving the behavior of AESER on other datasets and more fine-grained scoring rubrics as an open direction for future work.

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